

FORM VI MATHEMATICS

Time allowed: 2 hours

Exam date: 15th May 2002

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the right margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new 4-page examination booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each booklet.

Checklist:

- SGS Examination booklets required.

QUESTION ONE (Start a new examination booklet)

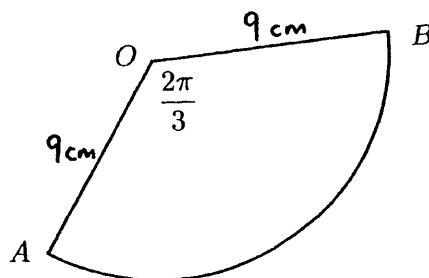
(a) Express $\frac{2\pi}{3}$ radians in degrees.

Marks

1

(b)

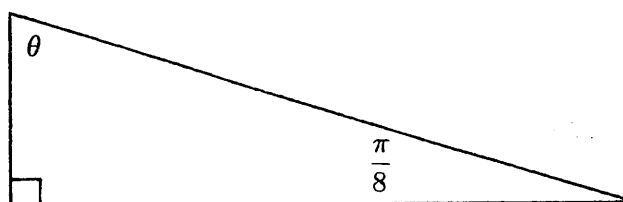
1



In the diagram above, AOB is a sector of the circle subtending an angle $\frac{2\pi}{3}$ at the centre O . Find the length of the arc AB .

(c)

1



In the diagram above, find θ in radians.

(d) Find $\log_e 3$, correct to four significant figures.

1

(e) Simplify:

(i) $\log_e \sqrt{e}$,

1

(ii) $1 + \log_e \frac{10}{e}$.

1

(f) Write down the range of $y = e^x$.

1

(g) Complete the following sentence:

1

"When $0 < x < 1$, the values of the function $f(x) = \log_e x$ are always"

(h) Write down the radius and the centre of the circle $(x - 3)^2 + y^2 = 5$.

2

(i) A variable point $P(x, y)$ moves so that it is equidistant from the point $(3, 0)$ and the line $x = -3$. Write down the equation of the locus of P .

1

(j) Sketch the parabola $x^2 = 8y$, clearly showing the coordinates of the vertex and the focus, and the equation of the directrix.

3

QUESTION TWO (Start a new examination booklet)

(a) Differentiate each of the following with respect to x :

(i) $y = \frac{2}{x^3}$,

(ii) $y = \log_e 3x$,

(iii) $y = \frac{1}{e^x}$,

(iv) $y = (e^x - 2)^4$.

Marks

1

1

1

1

(b) Simplify $2 \log 2 - \log 3 + \log 12$, expressing your answer in the form $a \log b$.

1

(c) Solve $2 \log_m 4 + \log_m 9 = 2$, where $m > 1$.

2

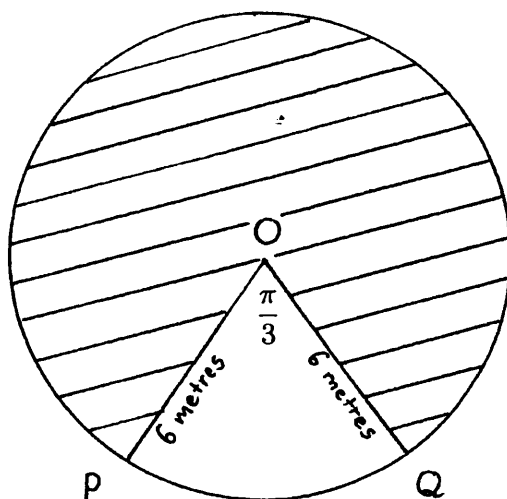
(d) (i) Find $\tan \frac{3\pi}{4}$.

1

(ii) Solve $\cos x = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$.

2

(e)



The circle above has centre O and radius 6 metres.

(i) Find the exact area of the major sector POQ .

2

(ii) Find the exact area of the minor segment cut off by the chord PQ .

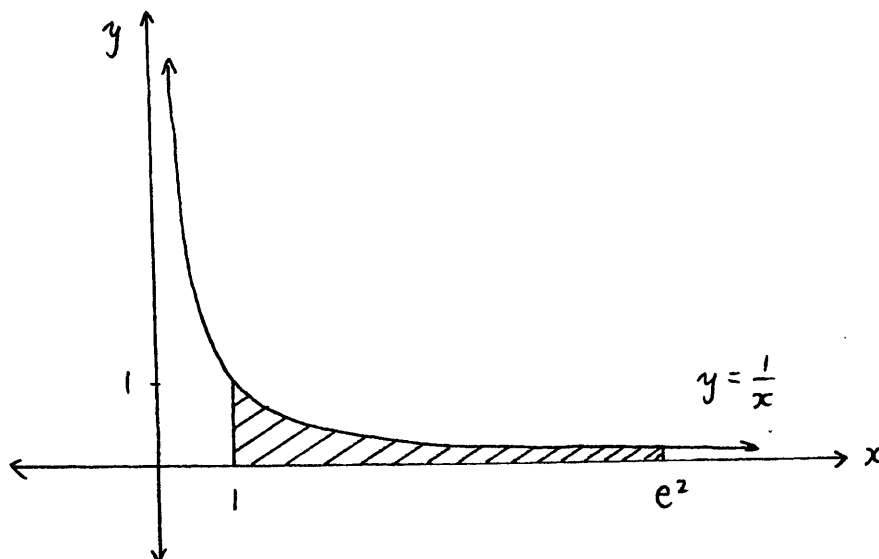
2

QUESTION THREE (Start a new examination booklet)

Marks

(a)

3



The diagram above shows the region bounded by the curve $y = \frac{1}{x}$, the x -axis, and the lines $x = 1$ and $x = e^2$. Find the area of the shaded region.

(b) Consider the curve $y = -x^3 + 3x^2 + 9x - 11$.

(i) Show that $\frac{dy}{dx} = -3(x-3)(x+1)$.

1

(ii) Find the coordinates of any stationary points and determine their nature.

4

(iii) Find the coordinates of any points of inflexion.

3

(iv) Sketch the curve, clearly showing the y -intercept and all stationary points and inflexions.

2

(v) For what values of x is the curve increasing?

1

QUESTION FOUR (Start a new examination booklet)

(a) Find the indefinite integrals:

Marks

(i) $\int e^{\frac{x}{2}} dx,$

1

(ii) $\int \frac{2x^2 - 4x}{x} dx,$

2

(iii) $\int (2x - 1)^{10} dx.$

1

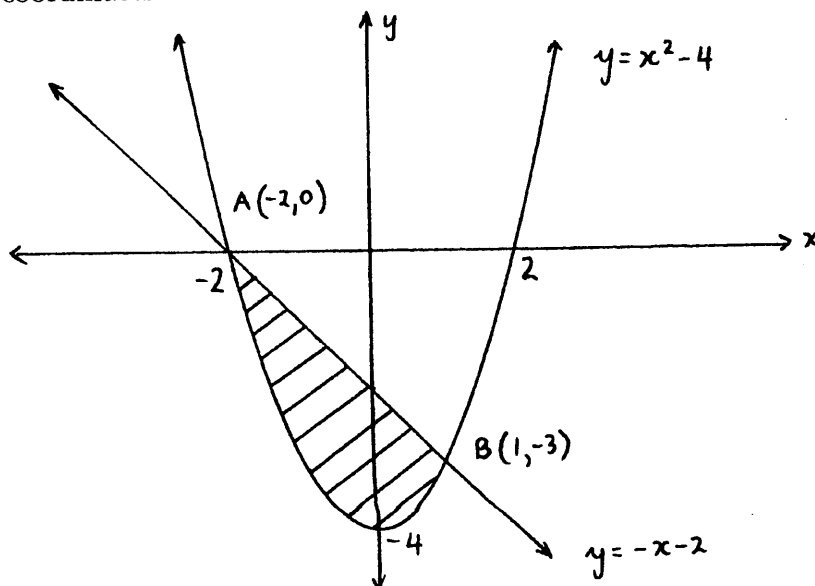
(b) Find the equation(s) of the locus of the point $P(x, y)$ such that the distance of P from the x -axis is always four times the distance of P from the y -axis. 2

(c) Consider the parabola with equation $4y = -x^2 + 4x - 16$.

(i) By completing the square, write the equation of the parabola in the form $(x - h)^2 = -4a(y - k)$. 1

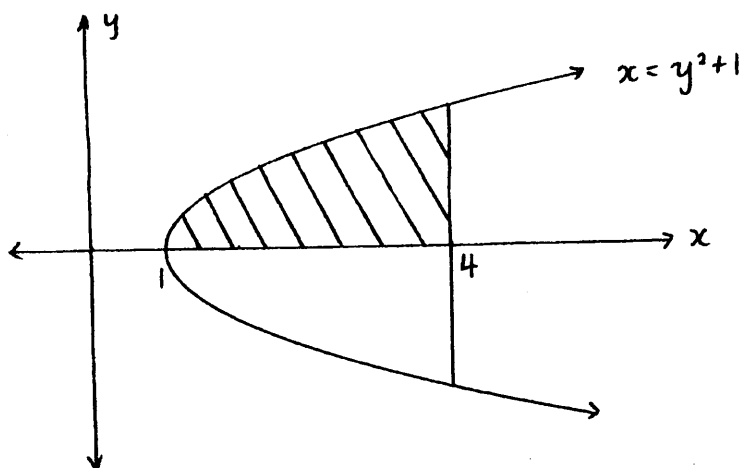
(ii) Find the coordinates of the focus. 1

(d)



The diagram above shows the region bounded by the line $y = -x - 2$ and the parabola $y = x^2 - 4$. The two functions intersect at $A(-2, 0)$ and $B(1, -3)$. Find the area of the shaded region. 3

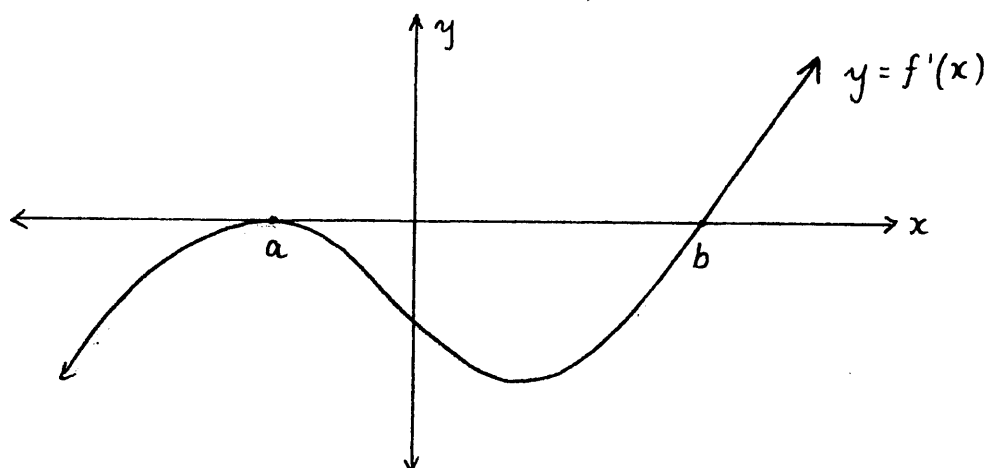
(e)



The diagram above shows the region above the x -axis bounded by the x -axis, the parabola $x = y^2 + 1$ and the line $x = 4$. Find the volume formed when the shaded region is rotated about the x -axis.

QUESTION FIVE (Start a new examination booklet)

(a)



The diagram above shows the graph of the derivative $y = f'(x)$ of a function $y = f(x)$. Marks

- (i) Write down the nature of the two stationary points on the graph of $y = f(x)$ at $x = a$ and at $x = b$. 2
 - (ii) Given that $f(0) = 0$, sketch a possible graph of $y = f(x)$. 2
- (b)
- (i) Show that the point $P(2, e)$ lies on the curve $y = e^{\frac{x}{2}}$. 1
 - (ii) Show the gradient of the tangent at P is $\frac{e}{2}$. 1
 - (iii) Show that the equation of the tangent to $y = e^{\frac{x}{2}}$ at P passes through the origin. 2
 - (iv) Show that the equation of the normal at P has equation $2x + ey - e^2 - 4 = 0$. 1
 - (v) Find the point Q where the normal crosses the y -axis. 1
 - (vi) Find the area of $\triangle POQ$. 1
- (c) Consider the curve $y = \frac{x^2}{e^x}$.
- (i) Show that $\frac{dy}{dx} = \frac{x(2-x)}{e^x}$. 1
 - (ii) Hence find the coordinates of the points on the curve where the tangent is horizontal. 2

QUESTION SIX (Start a new examination booklet)

Marks

- (a) The graph of $y = f(x)$ is known to have a maximum turning point at $(-1, 3)$. Given that $f''(x) = 12x + 4$, find the equation of the function $y = f(x)$. 4

- (b) (i) Show that $\frac{d}{dx}(x \log_e x) = \log_e x + 1$. 1

- (ii) Hence find the exact value of A , where 3

$$A = \int_1^4 \log_e x \, dx.$$

- (iii) Sketch the graph of $y = \log_e x$ and shade the region represented by A . 1

- (iv) Copy and complete the following table for the curve $y = \log_e x$, giving exact values for y : 1

x	1	2	3	4
y				

- (v) Use the trapezoidal rule with four function values to show that $A \doteq \log_e 12$. 2

- (vi) Find the second derivative of $y = \log_e x$ and hence explain why the approximation of A in part (v) is less than the exact area found in part (ii). 2

QUESTION SEVEN (Start a new examination booklet)

Marks

- (a) (i) Find $\lim_{x \rightarrow \infty} (1 - e^{2-x})$. 1
- (ii) Sketch $y = 1 - e^{2-x}$, showing the horizontal asymptote and any intercepts with the coordinate axes. 2
- (b) Consider the function $f(x) = e^x + e^{-x}$.
- (i) Show that $f(x)$ is an even function. 1
- (ii) Find $f(1)$, correct to three significant figures. 1
- (iii) Show that $f''(x) = f(x)$. 1
- (iv) Find any stationary points on $y = f(x)$ and determine each point's nature. 2
- (v) Explain why $y = f(x)$ is concave up for all real x . 1
- (vi) Sketch $y = f(x)$, clearly showing any intercepts with the coordinate axes and one other point on the curve. 1
- (c) (i) Write down the domain of $y = \ln(x - 2)$. 1
- (ii) The region bounded by $y = \ln(x - 2)$, the coordinate axes and the line $y = \ln 3$ is rotated about the y -axis. Show that the volume of the solid formed is $4\pi(3 + \ln 3)$ cubic units. 3

TCW

QUESTION 1

(a) 120° ✓

(b) $s = r\theta$

arc AB = $9 \times \frac{2\pi}{3}$

= 6π cm ✓

(c) $\theta = \frac{\pi}{2} - \frac{\pi}{8}$

= $\frac{3\pi}{8}$ ✓

(d) $\log_e 3 = 1.099$ (4 sig figs) ✓

(e) (i) $\log_e \sqrt{e} = \frac{1}{2}$ ✓

(ii) $1 + \log_e \frac{10}{e} = 1 + \log_e 10 - \log_e e$
 $= \log_e 10$ ✓

(f) $y > 0$ ✓

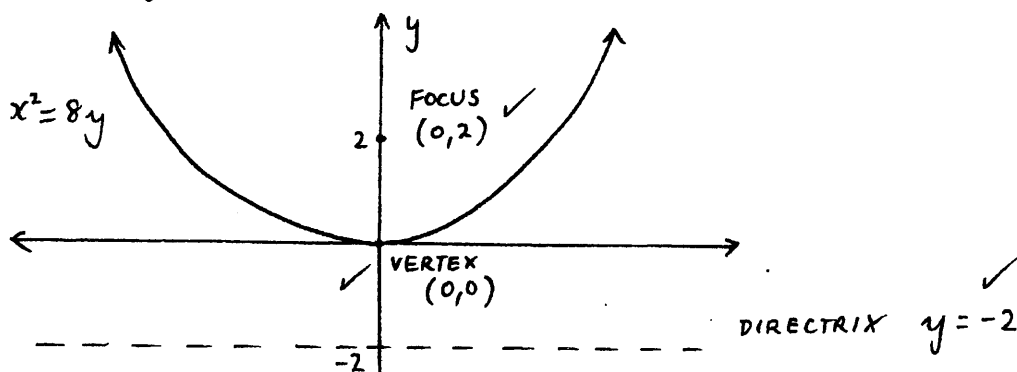
(g) negative ✓

(h) radius = $\sqrt{5}$ units ✓

centre (3,0) ✓

(i) $y^2 = 12x$ ✓

(j)



QUESTION 2

(a) (i) $y = 2x^{-3}$
 $y' = -6x^{-4}$
 $= -\frac{6}{x^4}$ ✓

(ii) $y = \log_e 3x$
 $y' = \frac{3}{3x}$
 $= \frac{1}{x}$ ✓

(iii) $y = e^{-x}$
 $y' = -e^{-x}$ ✓

(iv) $y = (e^x - 2)^4$
 $y' = 4e^x(e^x - 2)^3$ ✓

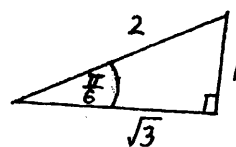
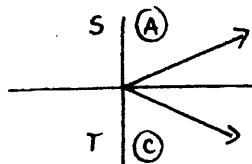
(b) $2 \log 2 - \log 3 + \log 12 = \log 4 - \log 3 + \log 4 + \log 3$
 $= 2 \log 4$ ✓

(c) $2 \log_m 4 + \log_m 9 = 2, m > 1$
 $\log_m 144 = 2$
 $m^2 = 144$
 $m = 12$ ✓

(d) (i) $\tan \frac{3\pi}{4} = -1$ ✓

(ii) $\cos x = \frac{\sqrt{3}}{2}$

$x = \frac{\pi}{6} \text{ or } \frac{11\pi}{6}$ ✓✓



(e) (i) $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 36 \times \frac{5\pi}{3}$
 $= 30\pi \text{ m}^2$ ✓

(ii) $A = \frac{1}{2} r^2 (\theta - \sin \theta)$
 $= 18 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$
 $= 6\pi - 9\sqrt{3} \text{ m}^2$ ✓

[NO PENALTY FOR UNITS]

QUESTION 3

$$\begin{aligned}
 (a) \quad A &= \int_1^{e^2} \frac{1}{x} dx \quad \checkmark \\
 &= \left[\log_e x \right]_1^{e^2} \quad \checkmark \\
 &= \log_e e^2 - \log_e 1 \\
 &= 2 \text{ units}^2 \quad \checkmark
 \end{aligned}$$

$$(b) \quad (i) \quad y = -x^3 + 3x^2 + 9x - 11$$

$$\begin{aligned}
 \frac{dy}{dx} &= -3x^2 + 6x + 9 \\
 &= -3(x^2 - 2x - 3) \quad \checkmark \\
 &= -3(x-3)(x+1)
 \end{aligned}$$

(ii) stationary points when $\frac{dy}{dx} = 0$

ie $x = 3, y = 16$ $\checkmark$
 $x = -1, y = -16$ $\checkmark$

x	-2	-1	0	3	4
y'	-15	0	9	0	-15

slope: $\quad \backslash \quad - \quad / \quad - \quad \backslash$ $\checkmark$

so $(3, 16)$ is a maximum turning point and $(-1, -16)$ is a minimum turning point. $\checkmark$

$$\begin{aligned}
 (iii) \quad \frac{d^2y}{dx^2} &= -6x + 6 \quad \checkmark \\
 \text{when } \frac{d^2y}{dx^2} &= 0 \\
 6x &= 6 \\
 x &= 1, y = 0 \quad \checkmark
 \end{aligned}$$

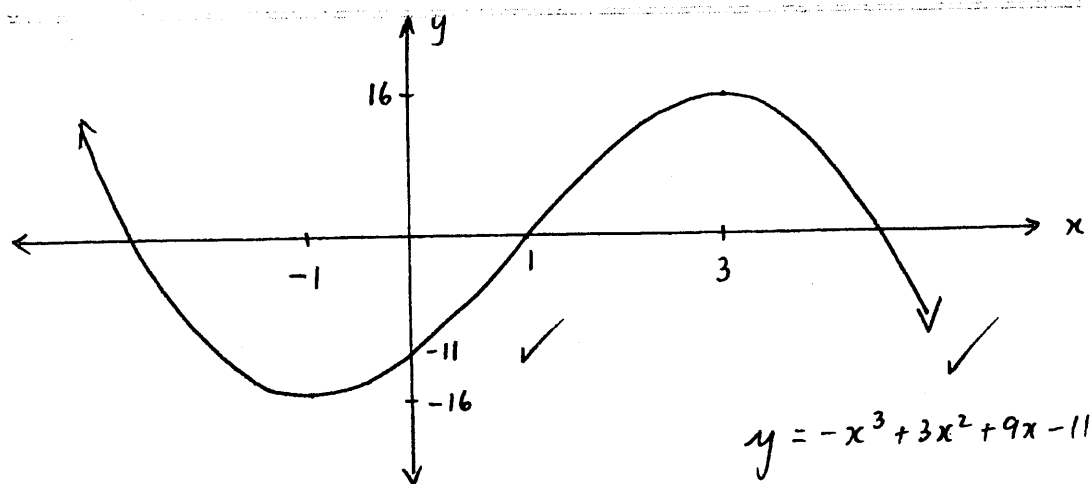
Test for a change in concavity:

x	0	1	2
y''	6	0	-6

$\cup \quad \quad \cap$ $\checkmark$

so $(1, 0)$ is a point of inflexion

(iv)



when $x=0$, $y=-11$

(v) increasing when $\frac{dy}{dx} > 0$

$$-3(x-3)(x+1) > 0$$

$$-1 < x < 3$$

✓

QUESTION 4

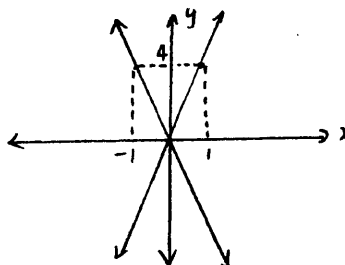
(a) (i) $\int e^{\frac{x}{2}} dx = 2e^{\frac{x}{2}} + c$ ✓

(ii) $\int \frac{2x^2 - 4x}{x} dx = \int 2x - 4 dx$ ✓
 $= x^2 - 4x + c$ ✓

(iii) $\int (2x-1)^{10} dx = \frac{1}{22} (2x-1)^{11} + c$ ✓

(b)

$y = 4x$ and $y = -4x$



✓✓

(c) (i) $4y = -x^2 + 4x - 16$
 $x^2 - 4x = -4y - 16$
 $x^2 - 4x + 4 = -4y - 12$
 $(x-2)^2 = -4(y+3)$ ✓

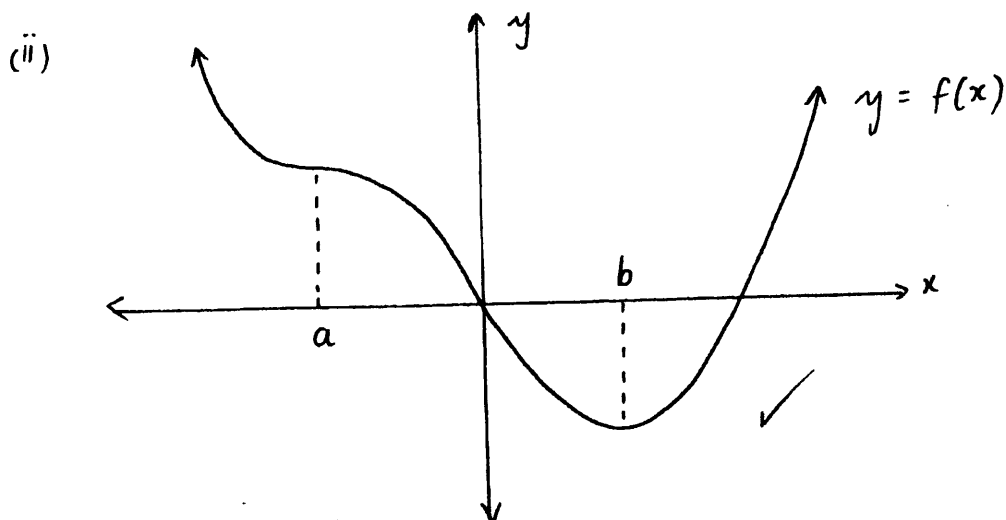
(ii) Focus = $(2, -4)$ ✓

(d) Area = $\int_{-2}^1 -x - 2 - x^2 + 4 dx$
 $= \int_{-2}^1 -x^2 - x + 2 dx$ ✓
 $= \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_{-2}^1$ ✓
 $= -\frac{1}{3} - \frac{1}{2} + 2 - \frac{8}{3} + 2 + 4$
 $= 4\frac{1}{2} \text{ units}^2$ ✓

(e) $V = \pi \int_1^4 (x-1) dx$ ✓
 $= \pi \left[\frac{x^2}{2} - x \right]_1^4$ ✓
 $= \pi \left(8 - 4 - \frac{1}{2} + 1 \right)$
 $= \frac{9\pi}{2} \text{ units}^3$ ✓

QUESTION 5

- (a) (i) Stationary point of inflexion at $x=a$. ✓
Minimum turning point at $x=b$. ✓



$$f(0) = 0$$

- (b) (i) $y = e^{\frac{x}{2}}$
when $x=2$, $y = e^{\frac{2}{2}}$
 $= e$ ✓
so $(2, e)$ lies on $y = e^{\frac{x}{2}}$

(ii) $y' = \frac{1}{2} e^{\frac{x}{2}}$
when $x=2$, $y' = \frac{1}{2} e^{\frac{2}{2}}$
 $= \frac{e}{2}$ ✓

so $\frac{e}{2}$ is the gradient of the tangent at $P(2, e)$

(iii) tangent at $P(2, e)$: $y - e = \frac{e}{2}(x - 2)$
 $2y - 2e = ex - 2e$

$$y = \frac{e}{2}x$$

when $x=0$, $y = \frac{e}{2} \times 0$
 $= 0$ ✓

so the tangent passes through the origin. ✓

(iv) gradient of normal at $P = -\frac{2}{e}$

normal at $P: y - e = -\frac{2}{e}(x - 2)$

$$ey - e^2 = -2x + 4$$

$$2x + ey - e^2 - 4 = 0$$

✓ as required

(v) when $x=0$, $ey = 4 + e^2$

$$y = e + \frac{4}{e}$$

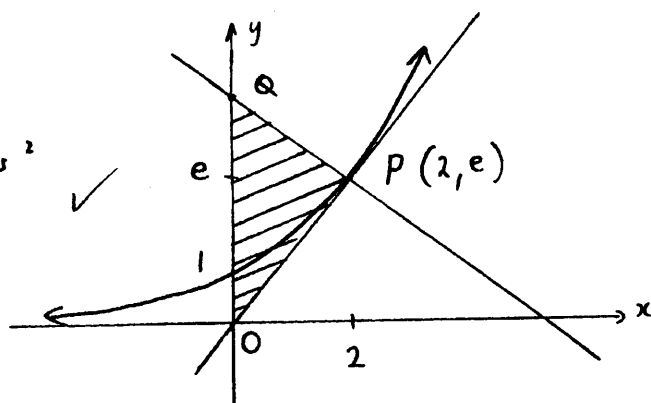
so Q is $(0, e + \frac{4}{e})$

(vi) Area = $\frac{1}{2} \times 2 \times (e + \frac{4}{e})$

$$= e + \frac{4}{e} \text{ units}^2$$

OR

$$\frac{4 + e^2}{e} \text{ units}^2$$



(c) (i) $y = \frac{x^2}{e^x}$

$$\frac{dy}{dx} = \frac{2x \times e^x - e^x \times x^2}{(e^x)^2}$$

by the quotient rule

$$= \frac{x e^x (2 - x)}{e^{2x}}$$

$$= \frac{x(2-x)}{e^x}$$

(ii) For a horizontal tangent, $\frac{dy}{dx} = 0$

$$x(2-x) = 0$$

$$x = 0 \text{ or } 2$$

$$y = 0 \text{ or } \frac{4}{e^2}$$

so the tangent is horizontal at $(0, 0)$ and $(2, \frac{4}{e^2})$.

QUESTION 6

(a)

$$f''(x) = 12x + 4$$

$$f'(x) = 6x^2 + 4x + C_1$$

$$0 = 6 - 4 + C_1 \quad \text{since } f'(-1) = 0$$

$$C_1 = -2$$

$$f'(x) = 6x^2 + 4x - 2$$

$$f(x) = 2x^3 + 2x^2 - 2x + C_2$$

$$3 = -2 + 2 + 2 + C_2 \quad \text{since } f(-1) = 3$$

$$C_2 = 1$$

$$\text{so } f(x) = 2x^3 + 2x^2 - 2x + 1$$

(b)

$$(i) \quad \frac{d}{dx} (x \log_e x) = 1 \times \log_e x + \frac{1}{x} \times x \quad \text{by the product rule}$$

$$= \log_e x + 1$$

(ii)

$$\int_1^4 \log_e x + 1 \, dx = [x \log_e x]_1^4$$

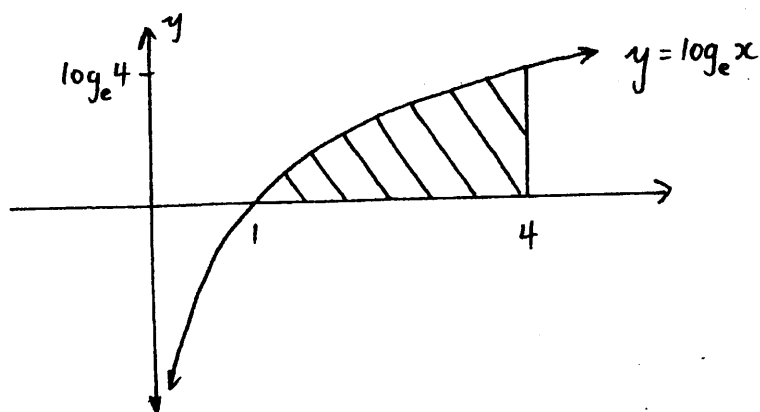
$$\int_1^4 \log_e x \, dx + \int_1^4 1 \, dx = [x \log_e x]_1^4$$

$$\int_1^4 \log_e x \, dx = 4 \log_e 4 - 0 - [x]_1^4$$

$$= 4 \log_e 4 - 3$$

$$\therefore A = 8 \log_e 2 - 3$$

(iii)



(iv)

x	1	2	3	4
y	0	$\log 2$	$\log 3$	$\log 4$

✓

(v)

$$A \doteq \frac{1}{2} (0 + \log 2) + \frac{1}{2} (\log 2 + \log 3) + \frac{1}{2} (\log 3 + \log 4)$$

$$\doteq \frac{1}{2} (2 \log 2 + 2 \log 3 + \log 4)$$

✓

$$\doteq \log 4 + \log 3$$

$$\doteq \log 12$$

✓

(vi)

$$y = \log_e x, \quad x > 0$$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}$$

✓

$$\frac{d^2y}{dx^2} < 0 \quad \text{for all } x$$

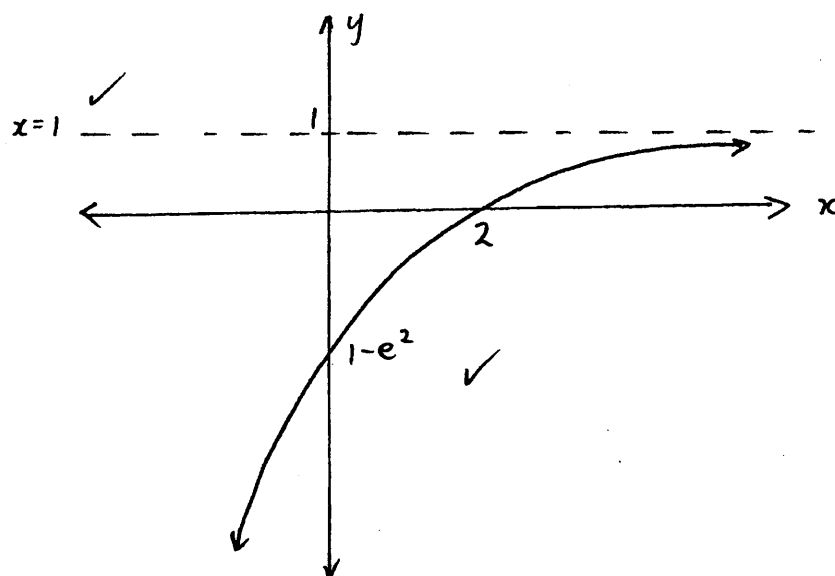
So $y = \log_e x$ is always concave down. Then the three trapezia applied in the trapezoidal rule all lie under the curve. Hence the approximation for A will be less than the exact value for A .

✓

QUESTION 7

(a) (i) $\lim_{x \rightarrow \infty} (1 - e^{2-x}) = 1$ ✓

(ii) when $x=0$, $y = 1 - e^2 \Rightarrow (0, 1 - e^2)$
 when $y=0$, $e^{2-x} = 1$
 $x = 2 \Rightarrow (2, 0)$



(b) (i) $f(x) = e^x + e^{-x}$
 $f(-x) = e^{-x} + e^{-(-x)}$
 $= e^{-x} + e^x$
 $= f(x)$ ✓

so $f(x)$ is an even function.

(ii) $f(1) = e + \frac{1}{e}$
 ≈ 3.09 (3 sig figs) ✓

(iii) $f(x) = e^x + e^{-x}$
 $f'(x) = e^x - e^{-x}$
 $f''(x) = e^x + e^{-x}$
 $= f(x)$ ✓

(iv)

$$\begin{aligned}
 f'(x) &= 0 \\
 e^x - e^{-x} &= 0 \\
 e^x &= \frac{1}{e^x} \\
 e^{2x} &= 1 \\
 2x &= 0 \\
 x &= 0, y = 2
 \end{aligned}$$

$$\begin{aligned}
 f''(0) &= 1 + 1 \\
 &= 2 \\
 &> 0
 \end{aligned}$$

so $(0, 2)$ is a minimum turning point.

✓

(v) Both e^x and e^{-x} are positive for all real x .

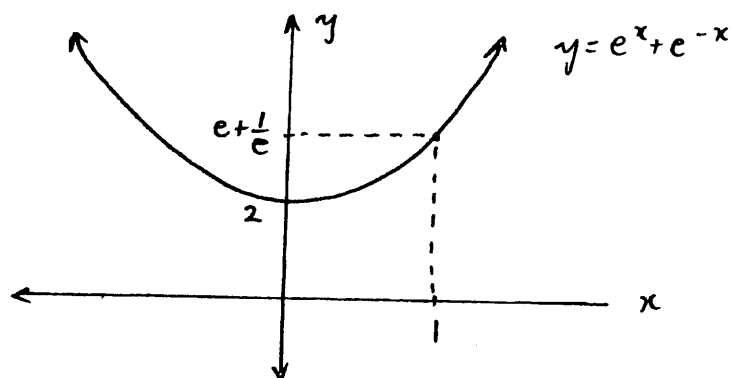
$$f''(x) = e^x + e^{-x}$$

$$> 0 \text{ for all real } x$$

so $y = f(x)$ is concave up for all real x .

✓

(vi)



[NOTE: MUST SHOW ONE OTHER POINT]

✓

(c) (i) Domain: $x > 2$ ✓

$$(ii) V = \pi \int_0^{\ln 3} (e^y + 2)^2 dy$$

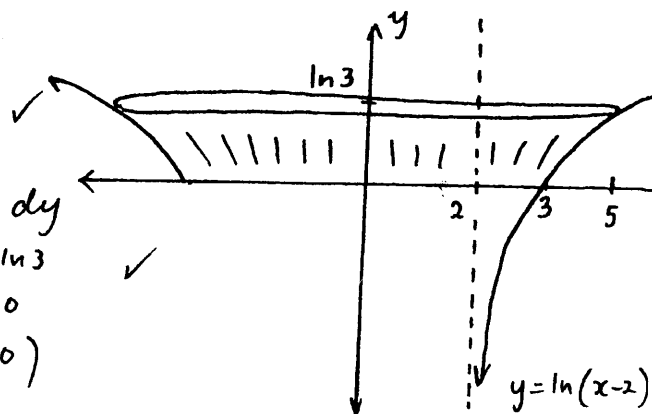
$$= \pi \int_0^{\ln 3} e^{2y} + 4e^y + 4 dy$$

$$= \pi \left[\frac{1}{2} e^{2y} + 4e^y + 4y \right]_0^{\ln 3}$$

$$= \pi \left(\frac{9}{2} + 12 + 4\ln 3 - \frac{1}{2} - 4 - 0 \right)$$

$$= \pi (12 + 4\ln 3)$$

$$= 4\pi (3 + \ln 3) \text{ units}^3 \quad \checkmark$$



14